

Hogfish

Species: *Lachnolaimus maximus*, Stock code: Hogfish - Florida Keys - East Florida

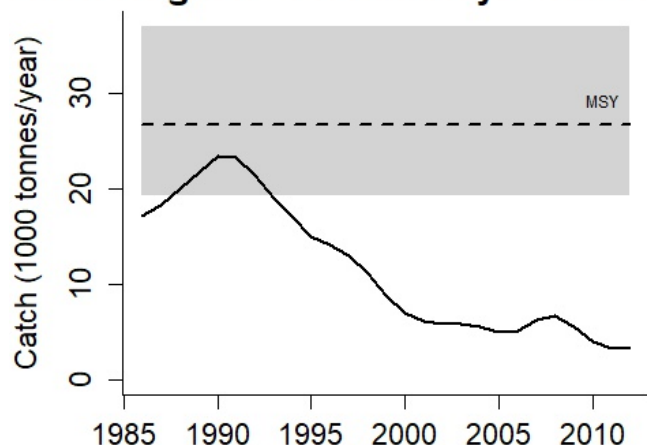
Region: Southeast Shelf

Marine Ecoregion: Florida Keys - East Florida

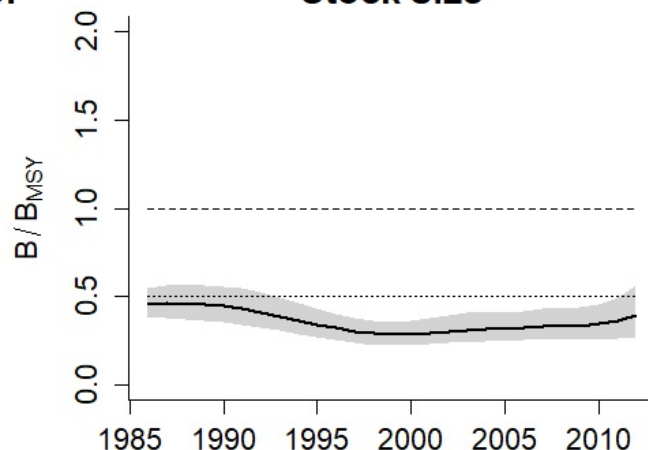
Reconstructed catch data used from years 1986 - 2012

For figure captions and method see <http://www.seaaroundus.org/cmsy-method>

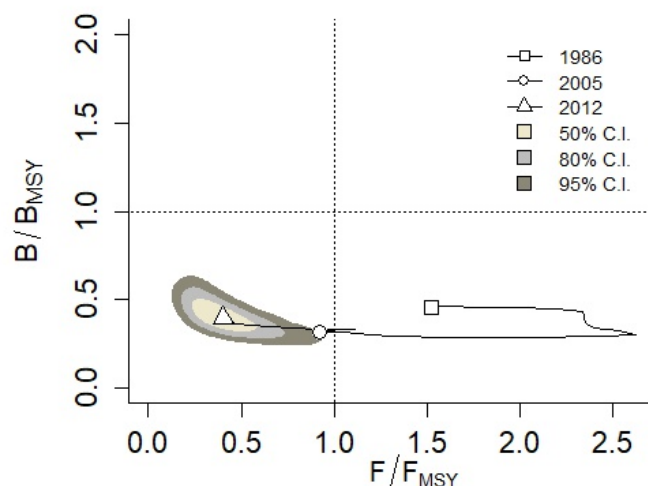
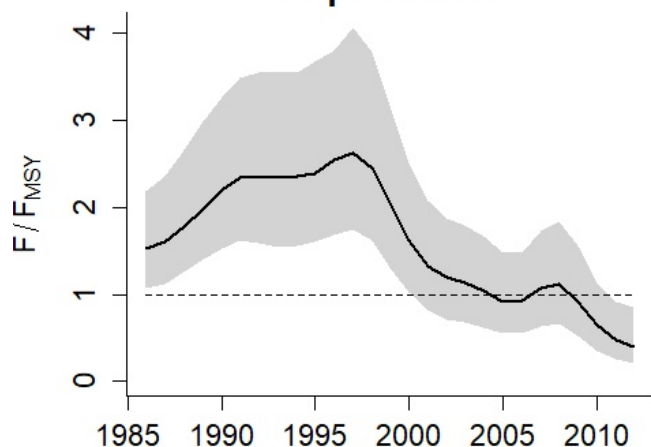
Catch Hogfish - Florida Keys - East Flori



Stock size



Exploitation



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Results for management (based on BSM analysis)

$F_{msy} = 0.0896$, 95% CL = 0.0547 - 0.147 (if $B > 1/2 B_{msy}$ then $F_{msy} = 0.5 r$)

$F_{msy} = 0.0708$, 95% CL = 0.0432 - 0.116 (r and F_{msy} are linearly reduced if $B < 1/2 B_{msy}$)

$MSY = 26.6$, 95% CL = 19.3 - 37; $B_{msy} = 296$, 95% CL = 174 - 495 (1000 tonnes)

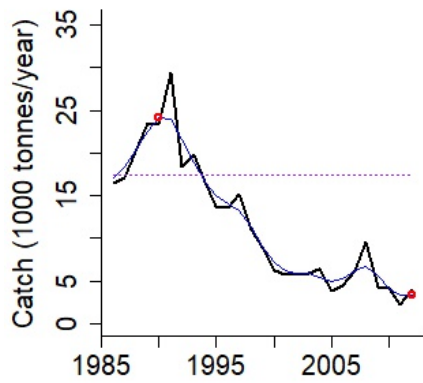
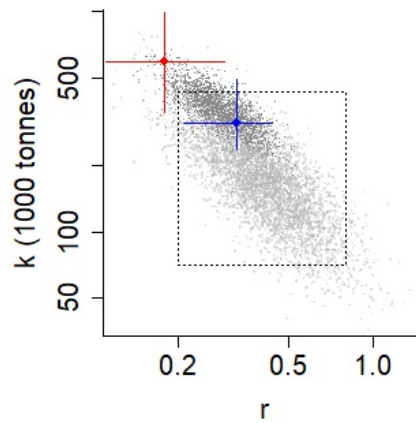
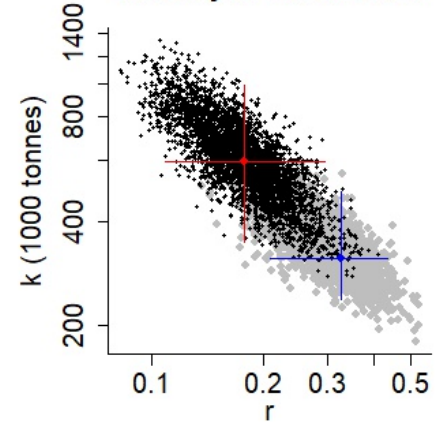
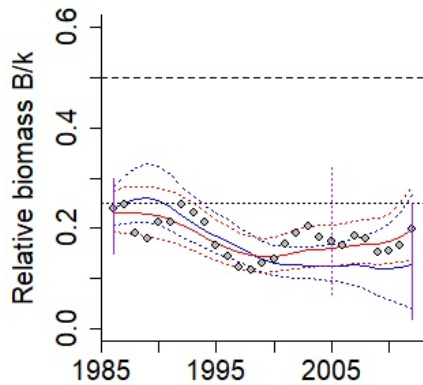
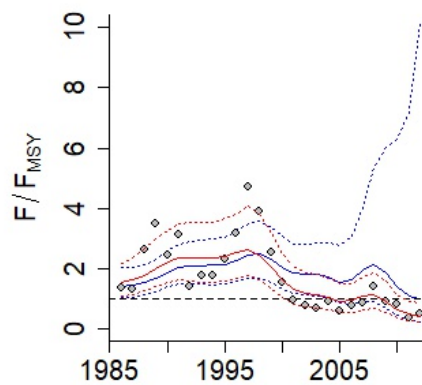
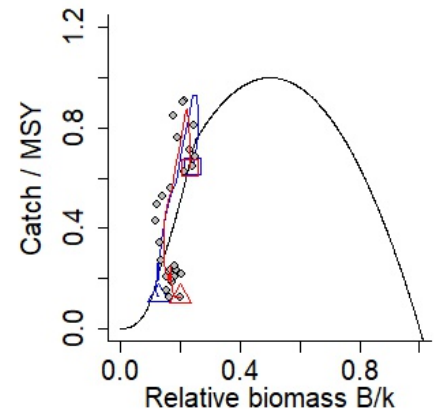
Biomass in last year = 118, 95% CL = 70.9 - 197 (1000 tonnes)

B/B_{msy} in last year = 0.395, 95% CL = 0.276 - 0.566

Fishing mortality in last year = 0.0285, 95% CL = 0.0159 - 0.0507

$F/F_{msy} = 0.401$, 95% CL = 0.209 - 0.845

Comment: NOAA Assessment model SS3. RF OK

A: Hogfish - Florida Keys - East Flor**B: Finding viable r-k****C: Analysis of viable r-k****D: Stock size****E: Exploitation rate****F: Equilibrium curve**

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Results of CMSY analysis conducted in JAGS

$r = 0.326$, 95% CL = 0.21 - 0.434; $k = 313$, 95% CL = 237 - 489 (1000 tonnes)

MSY = 25.5, 95% CL = 19.7 - 33 (1000 tonnes/year)

Relative biomass last year = 0.128 k , 95% CL = 0.0402 - 0.266

Exploitation $F/(r/2)$ in last year = 1.1

Results from Bayesian Schaefer model using catch and CPUE

$r = 0.179$, 95% CL = 0.109 - 0.294; $k = 593$, 95% CL = 347 - 990

r - k log correlation = -0.797

MSY = 26.6, 95% CL = 19.3 - 37 (1000 tonnes/year)

Relative biomass in last year = 0.128 k , 95% CL = 0.0402 - 0.266

Exploitation $F/(r/2)$ in last year = 0.58

$q = 3.41$, 95% CL = 2.12 - 5.55

Prior range of q = 1.14 - 28

Relative abundance data type = CPUE

Prior initial relative biomass = 0.15 - 0.3 expert

Prior intermediate relative biomass = 0.0678 - 0.323 in year 2005 default

Prior final relative biomass = 0.02 - 0.25 expert

Prior range for r = 0.2 - 0.8 default, prior range for k = 70.8 - 435 (1000 tonnes) default

Source for relative biomass:

NOAA Fisheries. (2020). Assessment Summary Data. Retrieved from www.st.nmfs.noaa.gov/stocksmart. 12/13/2020